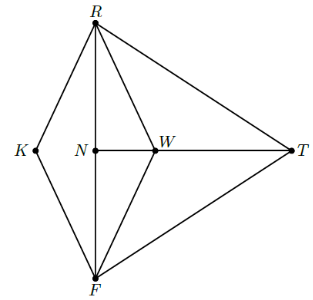
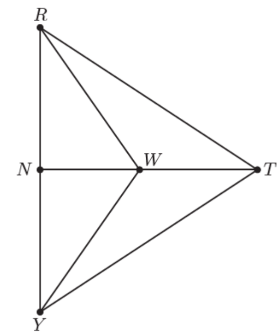
Geometry: Unit 5, Lesson 9

1. Quadrilateral $RTFK$ is shown, where $RN = 4.25$, $RT = 7.1$, $NT = 6.5$, $FR = 8.5$, $RW = 4$, $m\angle TWR = 115°$, $m\angle TFW = 26°$, $m\angle WRN = 33°$, and $m\angle RTW = 39°$. Given $\Delta RWF \cong \Delta RKF$ and $\Delta RWN \cong \Delta FWN$, what is the length of FW?



2. Triangle RTY is shown where $\overline{TN}$ is a perpendicular bisector of $\overline{RY}$ and point W is a point on $\overline{NT}$. How many different triangles are in triangle RTY?



3. Quadrilateral $BRDN$ is shown. Point W lies on $\overline{DR}$. How many different triangles are in quadrilateral $BRDN$?